DBMS Unit 4 - Transactions (Neha)

Transaction: a unit of work that must execute completely or not at all.

ACID properties
- Atomicity: all or nothing - handled by the recovery manager using the log (undo).
- Consistency: a transaction takes the database from one consistent state to another.
- Isolation: concurrent transactions do not see each other's intermediate results.
- Durability: once committed, changes survive crashes (redo from the log).

Transaction states: active -> partially committed -> committed; or failed -> aborted.

Concurrency problems: lost update, dirty read, unrepeatable read, phantom read.

Two-phase locking (2PL)
- Growing phase: the transaction may obtain locks but not release any.
- Shrinking phase: it may release locks but not obtain new ones.
- Strict 2PL keeps exclusive locks till commit, which avoids cascading rollbacks.

Shared (S) and exclusive (X) lock compatibility: S-S compatible, everything else conflicts.